Integration — Lesson 2 Test: Area Under a Graph = the Total

10 questions · show your work in the box · calculators allowed · every answer needs UNITS

Name: ______________________________ Date: ______________ Score: ______ / 10

1. A hose fills a barrel at a steady 7 liters per minute for 6 minutes. What is the area of the rectangle under the flow-rate graph, and how much water is in the barrel? 1 pt

2. A van drives at a steady 45 miles per hour from t = 2 hours to t = 6 hours. How far does it travel during that stretch? (Careful with the width!) 1 pt

3. A bicycle starts from rest and speeds up steadily to 12 m/s over 6 seconds, so its speed graph is a slanted line from 0. How far does it travel? 1 pt

4. A ball is dropped from a tall building. Its speed grows steadily from 0 to 20 m/s during the first 2 seconds. How far does it fall in those 2 seconds? 1 pt

5. A scooter already moving at 5 m/s speeds up steadily to 15 m/s over 4 seconds. Split the region under its speed graph into a rectangle plus a triangle and find the distance traveled. 1 pt

6. A graph shows rainfall rate in millimeters per hour on the y-axis and time in hours on the x-axis. The rate is a steady 3 mm/hour for 5 hours. Find the area under the graph, state its UNIT, and explain what the area means in real life. 1 pt

7. A race car accelerates from rest with a straight-line speed graph and covers exactly 48 meters in 8 seconds. What top speed does it reach at the end? 1 pt

8. A small business's earning rate climbs steadily from \$10 per day to \$20 per day over 5 days. Using rectangle + triangle, how much money does it earn in total? 1 pt

9. A car traveling 24 m/s brakes steadily to a complete stop over 4 seconds. Its speed graph slants DOWN from 24 to 0. How far does it travel while stopping? 1 pt

10. A tram cruises at 8 m/s for 5 seconds, then speeds up steadily from 8 to 16 m/s over 4 seconds, then cruises at 16 m/s for 10 seconds. Break the region under the speed graph into pieces and find the total distance. 1 pt